Safe Reinforcement Learning for Clinical Multi-Turn Agents Requires More Than Outcome Rewards

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Abstract

Reinforcement learning with verifiable rewards is increasingly used to improve language model reasoning, yet its effects on sequential decision procedures remain poorly understood. I study this problem in clinical multi-turn agents that must decide what information to request, when to stop gathering evidence, and which action to take. I introduce three controlled environments: medication refill triage, acute diagnostic work up, and dosing titration. Each environment includes hidden clinical states, query costs, exact adaptive oracles, and deterministic outcome verifiers. Across these tasks, outcome only group relative policy optimization often improves aggregate reward while producing undesirable procedures, including premature commitment, one question collapse, and failures on rare high risk cases. In diagnostics, assigning dense credit to questions that reduce uncertainty improves held out value and stabilizes high risk performance across three training seeds relative to outcome only reinforcement learning. Separating exploration from exploitation further increases high risk support, but at the cost of excessive information gathering on low acuity cases. Additional experiments show that the optimal strength of process credit is task dependent; reinforcement learning provides little benefit when supervised fine tuning has already exhausted the available headroom; and reinforcement learning does not reliably repair rare behavior gaps inherited from biased supervised data. These findings show that verifiable final outcomes are insufficient for training reliable multi turn agents. Reward design must account for the procedure used to reach an answer, and evaluation must distinguish adaptive information gathering from reward exploiting shortcuts. More broadly, the results identify reward geometry, initialization coverage, and task headroom as interacting determinants of whether reinforcement learning improves or degrades sequential reasoning in safety critical settings.

1 Introduction

Clinical decision making is often a sequential process of gathering evidence under uncertainty. A clinician evaluating acute chest pain may need to obtain an electrocardiogram, cardiac biomarkers, haemodynamic findings, and symptom history before deciding whether emergency treatment is warranted. A clinician reviewing a medication refill may need recent renal function, adverse effects, adherence, and contraindications before approving or modifying therapy. A clinician adjusting an anticoagulant must balance clotting risk against bleeding risk, often using information that is incomplete at the beginning of the encounter. In each setting, the quality of the decision depends not only on the final action, but also on which information was requested, how that evidence was interpreted, and when the decision maker chose to stop gathering.

Clinical language agents face the same sequential problem. An agent that acts from an incomplete prompt may miss a dangerous condition; an agent that asks every available question may create unnecessary delay and burden. The desired policy must therefore learn three related behaviours: what to ask, when enough information has been obtained, and which action to take. This makes clinical

multi turn interaction different from one shot medical question answering. The target is not merely a correct response to a fully specified vignette; it is a reliable decision procedure for reaching an action from an initially incomplete state.

Recent research has begun to evaluate medical language models as sequential decision makers rather than static question-answering systems. Nori et al. introduce the Sequential Diagnosis Benchmark (SDBench), in which agents iteratively request clinical findings and diagnostic tests before committing to a diagnosis [Nori et al., 2025]. Their MAI Diagnostic Orchestrator (MAI-DxO) demonstrates that structured, cost-aware inference-time orchestration can improve diagnostic efficiency without updating model parameters. This motivates a complementary learning question: whether reinforcement learning can induce effective information-gathering procedures within the policy itself, and under what conditions those procedures improve or degrade.

Reinforcement learning with verifiable rewards, or RLVR, is a natural candidate for training such procedures. Supervised fine tuning can teach an action schema, clinical protocols, and examples of desirable conversations. It does not directly optimize the tradeoff between requesting more evidence and committing to a decision. Reinforcement learning can, in principle, optimize that tradeoff. If the final action can be scored, and each additional question carries a cost, an agent can be rewarded for gathering useful information while avoiding unnecessary investigation. Recent work has extended RLVR from mathematics and code to medical reasoning [Chen et al., 2024, Zhang et al., 2025, Lin et al., 2025]; however, most medical RLVR evaluations still emphasize the correctness of a final answer rather than the quality of the sequential process used to obtain it.

This distinction creates an important safety problem. A terminal reward may assign a high score to the correct clinical action even when that action was reached through an unreliable procedure. A model may commit after one question, choose a broadly conservative action that performs adequately on common cases, or avoid a difficult decision by selecting an intermediate option. Such policies can improve average reward while becoming less reliable on rare, high risk cases. Conversely, a reward that strongly encourages information gathering may cause the model to ask too many questions, delay action, or escalate unnecessarily. Aggregate reward alone may therefore conceal clinically important changes in behaviour.

I study these learning dynamics in three controlled clinical environments: medication refill triage, acute diagnostic work up, and dosing titration. Each environment is formulated as an ask to commit decision process. The agent begins with an incomplete case, may request individual findings, receives deterministic answers from a simulator, and eventually commits to a clinical action. The environment includes a hidden clinical state, an explicit cost for each question, a hand specified utility function, and an exact adaptive oracle that identifies the optimal sequence of questions and actions. Cases are generated from the simulator rather than extracted from clinical records. The environments are therefore not intended as clinical validation benchmarks; their purpose is to isolate how reinforcement learning changes information gathering when the state, reward, and optimal policy can be measured exactly.

Figure 1 summarizes the sequential decision problem and illustrates how different reward designs can produce qualitatively different information-gathering procedures for the same underlying clinical state.

The central question is whether RL learns a better decision procedure, or merely a more effective way to exploit the reward. I first compare outcome only group relative policy optimization, or GRPO, with credit assignment GRPO, which gives additional training credit to questions that reduce uncertainty about the hidden clinical state. I then study a decoupled exploration and exploitation variant that separates information gathering from final commitment. Finally, I examine how the benefit of RL depends on supervised initialization, including a setting in which supervised fine tuning already performs strongly and a setting in which the supervised data contain a rare high risk coverage gap.

The results show that a verifiable terminal reward is not sufficient to produce a reliable multi turn policy. Plain GRPO often increases average reward by shortening the interaction, collapsing toward one question policies, or choosing actions that perform adequately on common cases while failing rare emergencies. In the diagnostics task, dense credit for uncertainty reducing questions improves held out value and stabilizes high risk performance across three training seeds. Separating exploration from exploitation produces the strongest high risk support, but it also increases information gathering on low acuity cases. The dosing experiments show that the strength of process credit acts as a task

dependent control parameter; too little credit permits under gathering, while too much encourages excessive gathering. The supervised initialization experiments reveal a further limitation: RL adds little when supervised fine tuning has already exhausted the available headroom, and it does not reliably repair a rare high risk coverage gap even when the missing action remains possible to sample.

The primary conclusion is therefore not that RL reliably improves clinical agents. Rather, RL can improve, preserve, or degrade the information gathering procedure depending on the task, initialization, and reward structure. Adaptive headroom is required because asking questions must offer an advantage over a fixed action. Useful behaviour must also be represented in the initial policy because policy optimization cannot reliably amplify trajectories that are never generated. Finally, the reward must assign credit to the procedure itself; otherwise, outcome optimization can favour shortcuts that score well without gathering the evidence needed for safe decisions.

These findings have implications for future clinical decision support systems. A model should not be considered safer merely because its average verifier score improves. Evaluation should also examine whether the model obtained the findings needed to justify its action, whether it committed prematurely, whether it overinvestigated low risk cases, and whether performance was preserved on uncommon but consequential scenarios. In real clinical systems, these concerns would extend to incomplete records, noisy patient responses, clinician disagreement, workflow burden, and uncertainty that cannot be represented by a deterministic verifier. The present environments simplify those factors, but they expose a basic risk that will remain relevant in more realistic systems: reinforcement learning can alter the decision procedure in clinically important ways even when final answer metrics appear to improve.

I make three main contributions. First, I introduce a controlled framework for studying clinical multi turn agents as ask to commit decision processes with exact verifiers, exact adaptive oracles, and separate evaluation of high risk cases. Second, I show that outcome only GRPO can learn shortcut procedures even when the task has measurable value of information and the initial model can sample useful behaviour; dense credit assignment improves and stabilizes the main diagnostics result. Third, I characterize several limits of RL post training: stronger exploration trades high risk performance against low acuity burden, process credit must be calibrated to the task, strong supervised models may leave little useful headroom, and RL does not reliably repair coverage gaps inherited from biased supervised data.

More broadly, this work frames reinforcement learning for clinical multi turn agents as procedure learning rather than final answer optimization. Safe post training requires more than a verifiable outcome; it requires a task in which adaptive gathering has value, an initialization that exposes useful behaviour, a reward that assigns appropriate credit to intermediate decisions, and an evaluation that audits rare failures rather than relying on aggregate performance alone.

2 Background

Verifiable reward learning for reasoning and medicine. Reinforcement learning with verifiable rewards has been most successful when the quality of an output can be checked directly. In mathematics, a proposed answer can be compared with a known solution; in programming, generated code can be executed against tests [Shao et al., 2024]. Recent work has extended this approach to medical reasoning by rewarding answers that agree with references, structured rubrics, or rule based evaluators [Chen et al., 2024, Zhang et al., 2025, Lin et al., 2025]. These studies suggest that verifiable feedback can improve medical reasoning, but they largely treat the model response as a completed product. Many clinical decisions instead emerge through a sequence of information requests, interpretations, and actions. For these problems, the object being optimised is not only the final answer; it is the procedure used to reach that answer.

This distinction is especially important in clinical settings. A correct medication decision may depend on whether renal function, allergies, adverse effects, or interacting therapies were checked first. An appropriate emergency disposition may depend on whether the agent identified and obtained the findings needed to justify escalation. A verifier that scores only the terminal action can therefore reward a policy that reaches the correct answer without following a procedure that would remain reliable when the case changes. Verifiability makes the learning signal measurable, but it does not by itself make the learned behaviour clinically appropriate.

Relation to RLHF and preference based clinical alignment. Not every clinically important property can be represented by a deterministic verifier. Communication quality, uncertainty expression, shared decision making, and preference-sensitive tradeoffs may require human judgement. Clinical post-training has therefore also explored preference-based alignment. Savage et al. compare supervised fine-tuning with direct preference optimisation across clinical tasks [Savage et al., 2025], while Dou et al. use diagnostic-process feedback to construct preferences for multi-turn medical dialogue [Dou et al., 2024]. These approaches complement the analytically specified reward setting studied here.

I study a complementary part of clinical alignment. In my environments, the hidden state, the cost of requesting information, and the utility of the final action can be specified analytically. This permits exact comparison with an adaptive oracle and makes it possible to identify whether reinforcement learning improves the decision procedure or exploits the scoring rule. I use RLVR because it provides this experimental control, not because deterministic rewards can represent every aspect of clinical quality. The distinction also matters when interpreting learned process supervision: feedback from another language model is a learned judgement signal, rather than a strictly verifiable reward, and it may introduce the preferences or biases of the supervising model.

The broader safety concern applies to both settings. If a policy can exploit a clean analytic verifier, it may also exploit a learned reward model. Preference based clinical alignment should therefore be evaluated not only through average reward or final answer accuracy, but also through audits of premature commitment, unsupported escalation, unnecessary information gathering, and performance on uncommon high consequence cases.

Clinical multi turn agents. Clinical multi turn agents interact with patient states that are incomplete at the beginning of an encounter. They must choose which question to ask next, update their decision after receiving the answer, and determine when further information is unlikely to change the appropriate action. This differs from medical question answering, where the relevant findings are usually presented in a complete vignette and the model produces a single response.

Recent medical dialogue agent research supports this sequential framing. For example, DoctorAgent RL uses a doctor agent that interacts with a language model patient agent and receives feedback for diagnostic accuracy, information acquisition, and protocol compliance [Feng et al., 2026]. Such work moves medical language models closer to proactive consultation, where the agent must elicit missing evidence rather than wait for a fully specified case. My focus is narrower and more controlled. I use deterministic simulators to isolate how reinforcement learning changes three parts of the policy: which information the agent requests, when it stops requesting information, and which action it takes after gathering evidence.

SDBench and MAI-DxO provide a complementary approach to sequential clinical reasoning [Nori et al., 2025]. SDBench evaluates diagnostic accuracy and investigation costs as agents acquire clinical information over multiple turns, while MAI-DxO uses inference-time orchestration to structure hypothesis refinement and test selection. Unlike this prompted orchestration approach, I investigate how reinforcement learning changes the information-gathering policy itself. My controlled environments expose exact hidden-state dynamics, deterministic outcome verifiers, and adaptive oracle policies, enabling direct measurement of premature commitment, unnecessary investigation, and high-risk failures. These simplifications permit a more controlled study of learning dynamics, although they do not reproduce the clinical realism of SDBench.

This controlled setting represents a simplified analogue of several realistic clinical workflows. In refill triage, an agent may need to check symptoms, adherence, recent laboratory values, and contraindications before continuing or modifying treatment. In acute assessment, it may need to obtain red flag findings before deciding whether emergency intervention is warranted. In dosing titration, it may need to balance competing treatment risks before increasing, reducing, or withholding therapy. These examples motivate the decision structure, although the synthetic environments do not model the noise, ambiguity, workflow constraints, or clinician disagreement present in actual care.

Reinforcement learning shapes procedures, not only answers. Post training can alter how a language model allocates inference time and organises intermediate reasoning, even when evaluation ultimately concerns a final output. SPIRAL, for example, trains a model to coordinate search, multiple reasoning traces, and aggregation, thereby optimising a broader inference procedure rather than a

single generated answer [Hamid et al., 2026]. My setting studies a different procedure: clinical gather then commit behaviour. The model must request relevant findings, use the revealed evidence, and commit only when additional information is no longer worth its cost.

This procedural view changes how performance should be interpreted. Two agents may choose the same final action but differ substantially in reliability. One may have gathered the findings that support the decision; the other may have guessed correctly after a single question. Similarly, an agent that asks every possible question may eventually choose the correct action while imposing unnecessary delay and burden. Final action accuracy alone cannot distinguish these policies. Evaluation must also measure how the action was reached.

Why dense credit matters for clinical information gathering. A terminal outcome reward provides feedback only after the dialogue has ended. When a model asks several questions before committing, that reward does not identify which question changed the decision, which question reduced uncertainty, or which question was unnecessary. Plain GRPO therefore faces a credit assignment problem. A successful trajectory may contain useful and wasted actions, yet every token in the trajectory is updated from the same final outcome. This can encourage simpler shortcuts, particularly when asking fewer questions reduces the explicit interaction cost.

Prior work on mathematical reasoning has distinguished final-answer correctness from the correctness of intermediate steps. Uesato et al. investigated process- and outcome-based feedback [Uesato et al., 2022], followed by work on human-labelled process supervision [Lightman et al., 2023] and automatically constructed process feedback for verification and reinforcement learning [Wang et al., 2024]. Related approaches to multi-step optimization include OREO, which jointly learns a policy and value function from offline trajectories [Wang et al., 2025], and SWiRL, which decomposes tool-use trajectories into action-level subtrajectories optimized using learned feedback [Goldie et al., 2025]. On-policy self-distillation instead provides token-level teacher supervision on student-generated trajectories [Zhao et al., 2026].

My approach instead assigns analytic information-gain credit to observable clinical information requests, without requiring a learned reward model or supervision of free-form reasoning traces. This credit is accumulated within the training reward for on-policy GRPO over complete dialogues, rather than used to optimize individual subtrajectories.

The process credit is used only during training. Every policy is evaluated with the same terminal utility and question cost, without the additional shaping signal. Any improvement must therefore appear as better gather then commit behaviour under the original task objective. This separation between training credit and evaluation reward is central to the experimental design. It allows me to test whether dense credit teaches a more reliable procedure, rather than merely increasing the shaped reward that the model was trained to optimise.

3 Experimental setup

I study clinical multi turn agents in a controlled ask to commit setting. Each episode begins with an incomplete clinical case. The agent may request missing findings, receive answers from a deterministic simulator, and then commit to one clinical action. Requesting information carries a small cost; the agent must therefore balance the value of additional evidence against delay and unnecessary questioning. The desired behaviour is not to ask as little or as much as possible. It is to obtain the information needed to support an appropriate action, then stop.

This design captures a simplified version of several clinical workflows. A refill agent may need recent renal function or adverse effect history before continuing a medication. An acute assessment agent may need red flag findings before choosing emergency treatment. A titration agent may need to compare treatment benefit with bleeding or toxicity risk before changing a dose. The environments remove much of the ambiguity of real care, but preserve the sequential decision problem that the experiments are intended to study.

Figure 1 summarizes the three clinical tasks, the supervised and reinforcement learning stages, the reward variants, and the common held out evaluation procedure.

Environment family. All three tasks instantiate the same partially observed decision process. A hidden clinical state, denoted by μ, represents the underlying patient condition. In refill triage and diagnostics, μ is a latent severity level. In titration, it is a joint clot and bleed risk state. Each case also contains a vector of binary clinical findings. Their values are sampled from task specific prevalences, then held fixed throughout the episode. The agent observes only findings that are initially provided or later requested.

At each turn, the policy emits one structured JSON action. It may either *ask* about a single finding or *commit* to a terminal clinical action. An ask is answered by the simulator; a commit ends the episode. Episodes are limited by a task specific turn budget, which prevents unlimited questioning and makes failure to reach a decision explicit.

Each question has cost $c{=}0.03$. The terminal reward is

$$R = U[\mu][a] - c\, n_{\text{asks}},$$

where U is a hand specified utility matrix, a is the committed action, and n_{asks} is the number of questions requested. The utility matrix assigns different consequences to each action under each hidden state. A decision that is reasonable for a low risk patient may therefore receive a poor score when the true state is high risk.

The question cost makes the clinical tradeoff explicit. For example, if the committed action has utility 0.90 and the agent asks three questions, the final reward is $0.90 - 3(0.03) = 0.81$. Asking an additional useful question is worthwhile only when the information is expected to improve the final decision by more than its cost. Asking a repeated or irrelevant question lowers reward without changing the patient state.

Exact adaptive oracle. Because the simulator, state prior, finding probabilities, question cost, and utility matrix are known, I can compute an exact adaptive oracle for every task. The oracle chooses the next question by considering how each possible answer would change the optimal action. It stops when no available question has enough expected value to justify its cost.

This oracle provides a target for the complete decision procedure, not only the terminal action. It identifies which findings should be requested, when gathering should stop, and which action should follow from the revealed evidence. A sampled trajectory is considered *oracle grade* when its realised reward is within EPS$=0.03$ of the oracle value. The tolerance equals the cost of one question, so a policy may differ from the oracle by one marginal request without being treated as a substantively different procedure.

Clinical tasks. I instantiate the framework in three stylized tasks that represent distinct forms of clinical uncertainty.

Refill triage models a medication continuation decision. The agent may need to ask about renal decline, dehydration, adverse effects, contraindications, or other medication safety findings before approving, modifying, denying, or escalating a refill. The two medication families are ACE inhibitor or angiotensin receptor blocker therapy and nonsteroidal anti inflammatory drug therapy.

Acute diagnostics models the work up of chest or abdominal presentations. The agent must decide whether to continue routine evaluation, choose an intermediate disposition, or initiate an emergency action. The task is deliberately constructed so that the emergency action requires several supporting findings. This exposes a clinically important shortcut: a policy can perform adequately on common cases by choosing an intermediate action while failing uncommon emergencies.

Dosing titration models an ordered treatment decision under competing risks. The agent chooses among three treatment intensities while considering both clot and bleed risk. Too little treatment may leave the patient unprotected; too much may increase harm. The correct action therefore depends on the combination of findings, rather than on a single severity variable.

Case generation and data splits. The core experiments do not use an external clinical dataset. Instead, cases are generated on demand from the task simulators. Each generated case contains a latent state, a fixed set of finding values, a clinical family, and the associated oracle policy. This

Table 1: Clinical environment dimensions. "Findings" is the set of binary clinical features available for the agent to request. "Commit actions" is the number of terminal decisions available to the policy. All environments use a question cost of 0.03 and an oracle grade tolerance of 0.03. Diagnostics contains the largest finding space and is designed to reveal hedging and premature commitment on rare emergency cases.

Task	Findings	Commit actions	Families	Latent state	Turn cap
Refill triage	12	5	2 (ACEi/ARB, NSAID)	severity (none/mod/high)	10
Diagnostics	18	5	2 (chest, abdominal)	severity (none/mod/high)	12
Titration	6	3 (ordered)	1 (anticoagulation)	joint clot$\times$bleed (9 states)	10

approach provides exact labels and complete control over case prevalence; it does not reproduce the noise, missingness, or demographic complexity of clinical records.

RL training uses 300 generated cases per task. Cases are drawn from the hidden condition, $\rho{=}0$, in which no decision relevant findings are revealed at the start of the episode. Training cases use generation seed 12345. Evaluation cases use seed 100 and are generated independently, so the training and held out sets are non overlapping samples from the same simulator distribution.

The main held out evaluation contains $n{=}60$ cases in the hidden condition. For the supervised coverage experiment, I increase the evaluation set to $n{=}200$ so that the uncommon high risk subgroup contains enough cases to support a separate analysis. The SFT warmups use 800 oracle or checklist trajectories generated with seed 7. These trajectories demonstrate the action format and desirable question sequences before reinforcement learning begins.

Training rollouts and evaluation sampling. During GRPO training, the policy generates a group of eight rollouts for each prompt. Relative rewards within that group determine the policy update. The model therefore learns from comparisons among several candidate ask and commit trajectories generated from the same initial case.

During held out evaluation, I sample $k{=}12$ trajectories per case. Following the coverage-based perspective of repeated sampling [Brown et al., 2024], I define support@k as the fraction of cases for which at least one sampled trajectory is oracle grade. This metric measures the availability of successful behaviour within the policy distribution, rather than its reliability in a single deployment trajectory. I therefore interpret support alongside realised value, mean number of questions, risk subgroup results, and audits of the trajectories themselves.

Unless otherwise stated, experiments use training seed 1. The main comparison between plain GRPO and credit assignment GRPO on diagnostics is repeated across three training seeds. The remaining experimental cells use one seed and are treated as exploratory results; this limitation is discussed in Section 6.

Safety focused evaluation. Aggregate reward can conceal clinically important failures because high risk cases represent only approximately 5–10% of the generated distribution. I therefore report that subgroup separately throughout the paper. I also examine the number and type of questions asked, premature commitment, one question collapse, unsupported escalation, hedging, and excessive gathering.

This distinction is necessary because two policies can receive similar aggregate scores for very different reasons. One may gather the findings required to support its action. Another may reach the same action through a fixed bias or a lucky guess. Likewise, a policy may improve high risk performance by escalating broadly, while creating unnecessary intervention on benign cases. The evaluation is designed to separate these behaviours rather than treating every high scoring trajectory as equally desirable.

Representative cases and oracle conversations for all three tasks are provided in Appendix B. Full SFT and GRPO hyperparameters are reported in Appendix A.

Scope and clinical interpretation. The environments are synthetic, stylized, and labelled by deterministic rules. They are not clinical validation benchmarks and should not be interpreted as evidence that any trained policy is suitable for patient care. The utility matrices are experimental design choices, not estimates of real clinical outcomes. Patient answers are deterministic, findings

are binary, and the simulator does not model documentation errors, ambiguous symptoms, conflicting records, patient preferences, or clinician disagreement.

These simplifications are intentional. By controlling the hidden state, available questions, question cost, utility function, and oracle policy, I can identify how reinforcement learning changes the information gathering procedure. The setup therefore supports conclusions about learning dynamics and reward design; it does not support conclusions about clinical efficacy. Its relevance to future decision support lies in the failure modes it exposes: premature action, unnecessary questioning, unsupported escalation, and apparently improved average performance that conceals deterioration on rare high consequence cases.

4 Methods

The experiments follow a common sequence. I first test whether adaptive information gathering can improve the task and whether the initial policy can produce useful trajectories. I then compare outcome only GRPO with two methods that change how credit is assigned to information gathering. Finally, I evaluate every trained policy using the same unshaped verifier. This separation ensures that a method cannot appear successful merely because its training reward is easier to optimise.

4.1 Pre RL eligibility gates

Before running policy optimisation, I apply two inexpensive eligibility tests. The first measures whether the task contains meaningful value of information. The second measures whether the initial policy can sample the behaviour that RL is intended to reinforce. These gates do not guarantee that RL will succeed; rather, they identify conditions in which RL has little plausible opportunity to help.

Gate 1: adaptive headroom. The first gate compares the exact adaptive oracle with the best policy that commits without requesting any information. I define the resulting best constant gap as

$$\text{BC gap} = V_{\text{oracle}} - \max_{a} \mathbb{E}_{\mu}[U[\mu][a]]\,,$$

where V_{oracle} is the expected value of the optimal adaptive policy and $U[\mu][a]$ is the utility of action a under hidden state μ.

A large positive gap indicates that asking questions can improve the decision over the best fixed action. A gap near zero indicates that the task offers little adaptive headroom; even an ideal information gathering policy performs only slightly better than committing immediately. In that setting, a small RL gain may reflect the structure of the task rather than a failure of the optimisation algorithm.

Gate 2: latent behavioural support. The second gate tests whether the initial policy can sample an oracle grade trajectory under stochastic decoding. For every held out case, I generate $k{=}12$ independent rollouts. A case is counted as supported when at least one rollout reaches a realised reward within EPS$=0.03$ of the exact adaptive oracle. The proportion of supported cases defines support@k.

This gate measures whether useful behaviour exists somewhere in the initial policy distribution. If support is zero for the target behaviour, policy gradient optimisation has no successful trajectory to amplify. In that situation, additional supervised training or improved data coverage is more appropriate than beginning RL directly from the current model.

Support@12 should not be interpreted as ordinary deployment reliability. It is a capability and exploration measure; a policy may produce one successful rollout among twelve while behaving poorly on most individual trajectories. I therefore interpret this gate separately from greedy performance, realised value, question burden, and high risk subgroup behaviour.

Gate 1 is analytic and depends only on the environment. Gate 2 is an inference only probe of the initial policy, either the base model or an SFT checkpoint. Neither gate uses the training rewards introduced below.

4.2 Model and optimisation setup

All methods use the text only `medgemma-4b-it` model [Saab et al., 2025]. I adapt the model using LoRA [Hu et al., 2022], which limits the number of trainable parameters while preserving the same base checkpoint across comparisons. Reinforcement learning uses group relative policy optimisation, or GRPO [Shao et al., 2024].

Training is fully on policy. For each clinical prompt, the model generates a group of candidate multi turn trajectories, receives a reward for each trajectory, and is updated according to relative performance within the group. The same model produces both information requests and terminal actions.

I use a disaggregated two node system. One node performs optimisation, while a separate vLLM server generates rollouts from the current policy. Simulator responses are inserted between model turns but are excluded from the policy gradient using an `env_mask`. Consequently, gradients are applied only to tokens generated by MedGemma; the model is not trained to imitate the deterministic patient simulator.

4.3 Outcome only GRPO control

Outcome only GRPO is the principal control. The policy receives the terminal clinical utility minus the accumulated cost of requesting information:

$$R_{\text{terminal}} = U[\mu][a] - c\, n_{\text{asks}}.$$

No intermediate question receives a separate reward. The entire trajectory is therefore updated from the final action, the true hidden state, and the total number of questions asked.

This is the most direct use of a clinical outcome verifier. It rewards an appropriate action and discourages unnecessary questioning. It also creates a credit assignment problem. If a trajectory contains several information requests, the model is not told which request changed the decision, which was irrelevant, or whether the final action succeeded despite inadequate evidence.

Outcome only optimisation can therefore favour procedures that are inexpensive but unreliable. A policy may commit after one question, repeatedly choose an intermediate action that performs reasonably on common cases, or escalate broadly without gathering the findings needed to justify the decision. These behaviours can improve average reward while weakening the intended gather then commit procedure.

4.4 Credit assignment GRPO

Credit assignment GRPO adds a dense training signal for questions that reduce uncertainty about the hidden clinical state. The terminal clinical objective is unchanged; the additional signal is intended to help the model identify which information requests contributed to the decision.

After each newly revealed finding, I update the posterior distribution over the hidden state. The information credit at turn t is the corresponding reduction in posterior entropy:

$$\Phi_t = H(p(\mu \mid \text{revealed}_{t-1})) - H(p(\mu \mid \text{revealed}_t))\,.$$

The training reward becomes

$$R_{\text{train}} = U[\mu][a] - c\, n_{\text{asks}} + \lambda \sum_t \Phi_t,$$

where λ controls the relative strength of information credit.

A question receives positive credit when its answer reduces uncertainty about the latent state. A repeated question, or a question that does not change the posterior, receives no information credit. The coefficient λ therefore controls how strongly the model values informative questions relative to their explicit cost.

This construction is motivated by potential based reward shaping [Ng et al., 1999]; however, I use it as an auxiliary process reward and do not claim that it is policy invariant under every task and training configuration. The coefficient can alter the preferred procedure. A weak coefficient may fail to prevent under gathering, while a strong coefficient may reward excessive questioning.

The distinction between training and evaluation is central. Information credit is used only during rollout training. It is not added to the utility matrix, the environment score, either pre RL gate, or any reported evaluation reward. A credit assignment policy must therefore improve the actual gather then commit trajectory to perform better under evaluation.

4.5 Decoupled exploration and exploitation GRPO

Credit assignment GRPO still optimises asking and committing within a single trajectory. The two behaviours may compete for the same gradient signal; asking more can increase information credit, while committing earlier can reduce question cost. To test whether a stronger separation improves high risk behaviour, I implement a decoupled exploration and exploitation variant inspired by DREAM [Liu et al., 2021].

This method is not an exact implementation of DREAM. It is a single model adaptation designed for the present ask to commit environments. The model uses one network and one LoRA adapter, with a mode token indicating whether the current update concerns exploration or exploitation.

During the exploration phase, the policy may request findings but may not commit. It is rewarded for gathering evidence that makes the hidden state recoverable. During the exploitation phase, the policy selects a terminal action using only the findings gathered during exploration. Its reward is the pure clinical utility of that action.

The two phases receive separate gradient updates, so the information gathering signal and the commitment signal are normalised within their respective phases. At evaluation time, the phases are composed into one dialogue. The policy first gathers information, then commits using the resulting transcript.

A hard information boundary prevents the exploitation phase from observing the privileged latent state or any finding that the exploration phase did not request. This guard is necessary to ensure that improved terminal decisions arise from gathered evidence rather than hidden simulator information.

The method tests whether stronger exploration improves rare high risk decisions. It also exposes the opposite failure mode. A policy that is strongly rewarded for recoverability may gather more information than is clinically necessary, particularly in benign or low acuity cases.

4.6 SFT followed by RL

Practical post training pipelines generally apply supervised fine tuning before reinforcement learning. In this setting, the relevant question is not whether RL can teach the task from the base model; it is whether RL improves behaviour that the supervised model has not already learned.

I study two forms of SFT initialization. In the refill triage task, RL begins from a light protocol teaching warmup. The supervised trajectories demonstrate the structured action format and common medication safety procedures. This experiment tests whether RL offers additional value after SFT has already learned much of the target behaviour.

In the diagnostics task, I construct a coverage biased SFT checkpoint by reducing the proportion of high risk oracle trajectories in the supervised data. This creates an initialization with strong aggregate performance but a specific rare case weakness. I then measure whether the policy can still sample the missing high risk behaviour and whether credit assignment or decoupled GRPO can repair the coverage gap.

These experiments separate two limitations. A well covered SFT policy may leave little adaptive headroom for RL; the supervised model already performs strongly on the measured objective. A biased SFT policy creates a different problem; the desired behaviour may be present only weakly or rarely in the policy distribution. Passing the latent support gate makes repair possible in principle, but does not guarantee that RL will preferentially reinforce the missing behaviour.

4.7 Common evaluation and guarded reporting

I evaluate all trained policies on held out cases using the pure terminal verifier:

$$R_{\text{eval}} = U[\mu][a] - c\, n_{\text{asks}}.$$

No information credit, exploration reward, or supervised trajectory score is used during evaluation. This common objective permits direct comparison among methods that were trained with different reward structures.

I report overall realised value, oracle grade support@k, and the mean number of questions requested. Where an experiment is replicated, I report variation across training seeds. I also analyse the high risk subgroup separately because uncommon cases can contribute little to the mean while accounting for the most consequential failures.

Aggregate metrics are not sufficient to determine whether the procedure improved. I therefore use a guarded report that audits the structure of each trajectory. The report includes:

- **Blind commitment:** the policy commits without requesting any decision relevant information.
- **One question collapse:** the policy repeatedly commits after a single request, even when the oracle requires additional evidence.
- **Premature commitment:** the policy acts before obtaining the findings needed to support the decision.
- **Unsupported escalation:** the policy selects a high acuity action without gathering the evidence required by the task.
- **Hedging:** the policy overuses an intermediate action that performs adequately on common cases but avoids the correct high stakes decision.
- **Over gathering:** the policy requests substantially more information than the oracle, particularly on benign or low acuity cases.
- **High risk regression:** performance deteriorates on the high risk subgroup even when aggregate value improves.

A policy may therefore receive a higher mean reward and still be classified as learning an exploitative procedure. For example, broadly selecting emergency treatment may reduce missed emergencies but does not constitute safer information gathering when the escalation is unsupported and benign cases are also over treated. Similarly, support@12 may remain high when one successful rollout is embedded within a distribution dominated by premature decisions.

The central evaluation criterion is whether the policy improves the complete procedure: it should gather the findings needed for the decision, avoid irrelevant requests, commit at an appropriate point, and preserve performance on rare high risk cases. Final outcome reward remains important, but it is interpreted together with the evidence gathering behaviour that produced it.

5 Results

The experiments reveal a consistent distinction between improving a terminal score and learning a reliable information gathering procedure. Outcome only GRPO can increase aggregate value while reducing the amount of evidence gathered and weakening performance on rare high risk cases. Credit assignment improves the main diagnostics comparison, but its effectiveness depends on the task and the strength of the information reward. Stronger separation between gathering and commitment improves high risk support, but increases questioning on low acuity cases. The SFT experiments show a further constraint; RL adds little when the supervised policy has already captured the available behaviour, and it does not reliably repair a rare coverage gap inherited from supervised training.

I present the results in six stages. First, I show that passing the pre RL gates does not ensure that outcome only GRPO will learn a safe procedure. Second, I compare outcome only GRPO with credit assignment across three diagnostics seeds. Third, I test whether separating exploration from

exploitation improves high risk behaviour. Fourth, I examine how the information credit coefficient controls under gathering and over gathering in titration. Fifth, I measure the remaining RL headroom after an SFT warmup. Finally, I test whether RL can repair a high risk coverage gap in a biased SFT policy.

Table 2 summarizes the final held out results. High risk value and high risk oracle grade support are the primary safety focused metrics. Overall value, overall support, and mean questions remain important, but none is sufficient in isolation. In several experiments, an apparently favourable aggregate score is produced by premature commitment, aggressive action selection, or excessive questioning rather than by adaptive evidence gathering.

Table 2: Final held out results across tasks and training methods. All policies are evaluated in the hidden condition using the pure verifiable reward and k=12 sampled rollouts per case. High risk value and high risk oracle grade support are the primary safety focused metrics. Overall value, overall support, and mean questions describe aggregate performance and interaction burden. Bold marks the strongest non exploitative high risk result within each task. Apparent support produced by a degenerate under gathering policy is marked EXPLOIT, because support@12 can remain high even when typical trajectories do not gather the evidence required for the decision. ‡Diagnostics plain GRPO and credit assignment GRPO are reported as means across three seeds; the corresponding standard deviations appear in Table 3. All other rows are single seed results.

Task	Method	overall val	overall supp	high val	high supp	asks
Diagnostics	base	$+0.251$	0.35	$+0.160$	0.50	2.5
	plain GRPO‡	$+0.285$	0.41	$+0.098$	0.33	1.8
	credit-assignment GRPO‡	$+0.313$	0.59	$+0.157$	$\mathbf{0.50}$	2.1
	decoupled GRPO	$+0.329$	0.42	$\mathbf{+0.248}$	$\mathbf{1.00}$	4.4
Titration	base	-0.022	0.52	$+0.396$	0.67	2.8
	plain GRPO	-0.012	0.57	$+0.627$	$1.00^{\dagger}$	1.4
	credit-assignment GRPO (λ=0.1)	-0.036	0.52	$\mathbf{+0.556}$	$\mathbf{1.00}$	2.7
Refill triage	base	$+0.386$	0.58	$+0.151$	0.00	2.2
	SFT warmup	$+0.318$	0.97	$\mathbf{+0.195}$	1.00	2.8
	plain GRPO	$+0.431$	0.97	$+0.032$	$1.00^{\dagger}$	1.3
	credit-assignment GRPO	$+0.415$	0.93	$+0.043$	$1.00^{\dagger}$	1.6

† High-risk support is a k-fold sampling artifact over a degenerate under-gathering policy (mean asks $\ll$ oracle), not adaptive gathering; the guarded report labels these EXPLOIT.

‡ Diagnostics plain GRPO and credit-assignment GRPO are 3-seed means; full mean$\pm$std values are reported in Table 3. All other rows are single-seed.

5.1 Passing the gates does not make outcome only GRPO safe

Diagnostics provides the clearest test of whether RL can improve an adaptive clinical procedure. The task passes both eligibility gates. Its best constant gap is $+0.161$, which shows that gathering information can outperform the best fixed action. The exact oracle requests approximately 3.9 findings per case. The base model also has nonzero support for oracle grade trajectories under stochastic decoding. The task therefore contains adaptive headroom, and the initial policy can produce at least some behaviour that RL could reinforce.

Outcome only GRPO does not reliably convert this opportunity into safer information gathering. Across three training seeds, overall value increases from $+0.251$ for the base model to $+0.285 \pm 0.027$. This aggregate improvement is accompanied by deterioration on the safety critical subgroup. High risk value falls from $+0.160$ to $+0.098 \pm 0.037$, while high risk support falls from 0.50 to 0.33 ± 0.29.

The variance across seeds is as important as the mean. One of the three plain GRPO policies reaches high risk support of 0.0, indicating complete loss of oracle grade high risk behaviour within twelve sampled trajectories per case. Mean questioning also falls from 2.5 at initialization to 1.82 ± 0.11, well below the oracle level. The policy is therefore improving its aggregate score partly by reducing the amount of evidence it gathers.

This result establishes the principal failure mode of outcome only optimisation. Passing the adaptive headroom and latent support gates identifies a setting in which RL could help; it does not ensure that the reward will favour the desired procedure. In diagnostics, outcome only GRPO improves the mean while making rare high risk behaviour less reliable.

5.2 Credit assignment improves and stabilizes diagnostics

Credit assignment changes the training signal but not the evaluation objective. During optimisation, the model receives additional credit for questions that reduce uncertainty about the hidden state. During held out evaluation, it receives only the original clinical utility minus question cost. An improvement therefore requires the shaped policy to perform better under the same terminal verifier as the control.

Across three seeds, credit assignment improves the principal diagnostics metrics. Overall value increases from $+0.285 \pm 0.027$ with outcome only GRPO to $+0.313 \pm 0.024$. High risk value increases from $+0.098 \pm 0.037$ to $+0.157 \pm 0.057$. Moderate support rises from 0.38 ± 0.18 to 0.67 ± 0.10. This moderate subgroup is particularly informative because additional findings frequently change the preferred disposition; the improvement is therefore consistent with more effective use of information rather than a fixed action bias.

The clearest benefit is the reduction in high risk instability. Plain GRPO has mean high risk support of 0.33 ± 0.29, including one seed with support of 0.0. Credit assignment maintains high risk support of 0.50 ± 0.00 across all three seeds. The mean does not exceed the base model value of 0.50, but the catastrophic seed failure observed under plain GRPO is absent.

These results support a limited but important conclusion. Dense information credit does not solve every high risk case, but it improves aggregate and subgroup value, increases moderate support, and makes the high risk result more consistent across the tested seeds. The improvement arises under the unshaped verifier, rather than from including information credit in the evaluation score.

Table 3: Diagnostics results in the held out hidden condition using the pure verifiable reward. Plain GRPO and credit assignment GRPO are reported as mean±sample standard deviation over three training seeds. Base and decoupled GRPO are single seed reference policies. Credit assignment improves overall value, high risk value, and moderate oracle grade support relative to the outcome only control. It also reduces high risk instability; one plain GRPO seed loses all high risk support, whereas credit assignment maintains support of 0.50 in each tested seed. "supp" denotes oracle grade support@12 and should be interpreted as sampled capability rather than single trajectory reliability.

Policy	overall value	high value	high supp	moderate supp	mean asks
base (untrained)	$+0.251$	$+0.160$	0.50	0.26	2.5
plain GRPO (control)	$+0.285 \pm 0.027$	$+0.098 \pm 0.037$	0.33 ± 0.29	0.38 ± 0.18	1.82 ± 0.11
credit-assignment GRPO (λ=0.2)	$\mathbf{+0.313 \pm 0.024}$	$+0.157 \pm 0.057$	$\mathbf{0.50 \pm 0.00}$	$\mathbf{0.67 \pm 0.10}$	2.08 ± 0.30
decoupled GRPO (best ckpt)	$+0.329$	$\mathbf{+0.248}$	$\mathbf{1.00}$	0.63	4.4

5.3 Decoupling increases high risk support but also question burden

Credit assignment still requires one policy trajectory to balance two competing objectives. Asking an additional question may improve information credit, while committing earlier avoids the explicit question cost. I therefore test a decoupled exploration and exploitation policy in which gathering and commitment receive separate updates.

At its best checkpoint, decoupled GRPO produces the strongest high risk result in diagnostics. Relative to credit assignment GRPO, high risk value increases from $+0.157$ to $+0.248$, and high risk support increases from 0.50 to 1.00. Overall value reaches $+0.329$.

The gain is accompanied by greater interaction burden. Mean questions rise to 4.4, above the base model, the credit assignment policy, and the oracle level of approximately 3.9. The increase is not distributed uniformly across clinical severity. In the checkpoint analysis, benign support falls from 0.48 to 0.04, indicating that the explorer continues requesting information in cases where earlier commitment would have been preferable.

Decoupled GRPO is therefore not a uniformly superior method. It moves the policy toward a conservative operating point that prioritizes high risk coverage at the cost of additional questioning and weaker low acuity efficiency. Such a tradeoff could be reasonable when a missed emergency has substantially greater cost than an unnecessary question, but the present experiment does not establish the clinical utility of that operating point. The result is also based on one training seed and should be interpreted as exploratory.

5.4 Information credit is a task specific control parameter

The titration task tests whether the information credit coefficient transfers across decision structures. The agent chooses among no therapy, reduced intensity, and full intensity while balancing clot and bleed risk. The task passes both gates, but its adaptive headroom is limited. The best constant gap is only $+0.075$. The exact oracle uses all three terminal actions; gathering changes the preferred action in 85% of cases, and the optimal priority changes between clot first and bleed first in 64% of cases. Base support@12 is 0.52 overall and 0.63 on the relevant subgroup.

The coefficient λ controls how strongly an informative question is valued relative to its cost. With no information credit, plain GRPO under gathers. Mean questions fall from 2.76 for the base model to 1.40, and the one question rate rises to 0.91. Although high clot support reaches 1.00, trajectory inspection shows that this result is produced by an aggressive low information policy rather than adaptive risk assessment. The guarded report therefore labels the behaviour EXPLOIT.

A weak coefficient, λ=0.03, does not correct the collapse. Mean questions fall further to 1.24, and the one question rate reaches 0.93. A strong coefficient, λ=0.2, produces the opposite failure. Mean questions increase to 4.10, substantially above the oracle level of approximately 2.77, while overall value falls to -0.091.

The intermediate setting, λ=0.1, produces the most calibrated gathering depth. This value follows the approximate relationship $\lambda \cdot \Delta H_{\text{ask}} \gtrsim c$, where a useful question reduces posterior entropy by approximately 0.33 nats and the question cost is 0.03. At λ=0.1, mean questions are 2.67, close to the oracle level. The one question rate falls to 0.62, and high clot support reaches 1.00 without the same degree of under gathering.

Calibrated gathering does not produce the highest overall value. At λ=0.1, overall value is -0.036, compared with -0.022 for the base model and -0.012 for plain GRPO. This result is consistent with the small best constant gap. The coefficient can improve the shape of the interaction, but a task with little adaptive headroom offers limited opportunity to convert better gathering into a higher terminal value.

Table 4: Titration results in the held out hidden condition using the pure verifier reward. The information credit coefficient controls the depth of gathering. Plain GRPO and λ=0.03 produce high one question rates and under gathering. At λ=0.2, the policy asks more than the oracle and loses aggregate value. The analytically motivated setting λ=0.1 places mean questioning close to the oracle level and preserves high clot support. The absence of an overall value gain is consistent with the small best constant gap of $+0.075$, which leaves limited adaptive headroom.

Policy	overall value	overall asks	one-ask rate	high-clot value	high-clot supp
base (untrained)	-0.022	2.76	0.74	$+0.396$	0.67
plain GRPO (control)	-0.012	1.40	0.91	$+0.627$	1.00
+ credit assignment (λ=0.03)	-0.010	1.24	0.93	$+0.825$	1.00
+ credit assignment (λ=0.1)	-0.036	2.67	0.62	$+0.556$	1.00
+ credit assignment (λ=0.2)	-0.091	4.10	0.31	$+0.446$	0.67

5.5 A strong SFT warmup leaves little safe RL headroom

Refill triage evaluates RL after a supervised protocol warmup. The question is no longer whether reinforcement learning can discover the task from the base model. It is whether RL improves a supervised policy that already demonstrates much of the desired medication safety procedure.

The SFT warmup reaches high risk value of $+0.195$ and high risk support of 1.00, with mean questioning of 2.8. Final plain GRPO and credit assignment GRPO also show high risk support of 1.00. This apparent preservation is misleading. The guarded report shows that support is generated by stochastic sampling around policies that gather substantially less information. Mean questions fall to 1.3 under plain GRPO and 1.6 under credit assignment. High risk value falls to $+0.032$ and $+0.043$, respectively.

The checkpoint trajectory reveals a short lived difference between the two RL methods. At the comparable early checkpoint, credit assignment maintains high risk value close to the SFT warmup, at $+0.205$, whereas the plain GRPO control has already fallen to -0.007. Continued optimisation removes this advantage, and both final policies converge toward cheaper under gathering procedures.

The result indicates that the supervised warmup had already captured much of the available high value behaviour. RL retained the ability to change the policy, but the remaining optimisation pressure did not correspond to useful clinical headroom. Reducing the number of questions was easier than improving an already strong safety procedure.

A small or negative gain beyond SFT should therefore not be interpreted as a general failure of reinforcement learning. In this experiment, it indicates that the supervised initialization had already solved much of the measured task and that the remaining terminal reward favoured efficiency over preserving the learned procedure.

5.6 Latent support does not guarantee repair of biased SFT

The final experiment tests whether RL can repair a rare coverage gap inherited from supervised training. I construct a diagnostics SFT dataset that under represents high risk emergency trajectories. The resulting policy performs well on aggregate metrics while retaining a clinically important weakness.

The biased SFT reaches overall value of $+0.337$ and overall support of 0.93, but its high risk value is -0.285. The discrepancy illustrates how aggregate performance can conceal a rare failure. Under severe bias, in which all high risk examples are removed, high risk support is 0.00 and the model never generates `emergent_tx`. Gate 2 therefore fails. Under milder bias, in which 30% of high risk examples remain, the model generates `emergent_tx` in 21% of sampled trajectories. Gate 2 passes, which shows that the missing action remains present in the policy distribution.

Passing the gate does not lead to successful repair. Starting from the biased SFT checkpoint, credit assignment GRPO reduces mean questions from 3.0 to 2.2. Overall support falls from 0.93 to 0.45, and high risk support falls from 1.00 to 0.14. The method therefore moves toward under gathering rather than restoring the missing high risk procedure.

Decoupled GRPO fails in the opposite direction. Mean questions rise to 4.3, while overall support falls to 0.41. High risk support reaches only 0.36, and benign support falls from 0.92 to 0.16. The explorer gathers more information, but does not recover the original supervised policy's strong low risk behaviour or reliably repair the high risk gap.

The experiment clarifies the role of the latent support gate. If the initial policy never samples the desired action, RL has no successful trajectory to amplify. Nonzero sampling is therefore a necessary condition for repair. It is not sufficient. When the desired behaviour is rare within an otherwise strong policy, GRPO may reinforce a more accessible neighbouring strategy, such as asking less or asking broadly, rather than selectively restoring the missing trajectory.

Coverage gaps must therefore be addressed through the supervised data, the reward distribution, targeted sampling, or a combination of these interventions. The presence of a rare action in stochastic rollouts should not be treated as evidence that standard RL will recover the associated clinical procedure.

Table 5: Attempted repair of a biased diagnostics SFT policy in the held out hidden condition, using n=200 cases and the pure verifier reward. The SFT initialization has strong aggregate and benign support but poor high risk value. Credit assignment GRPO shifts toward under gathering and sharply reduces high risk support. Decoupled GRPO shifts toward over gathering, damages benign support, and only partially recovers the high risk subgroup. Neither method repairs the original coverage gap. "supp" denotes oracle grade support@12.

Policy	overall val	overall supp	high val	high supp	benign supp	mean asks
biased SFT (init)	$+0.337$	$\mathbf{0.93}$	-0.285	$\mathbf{1.00}$	$\mathbf{0.92}$	3.0
+ credit assignment GRPO	$+0.236$	0.45	$+0.092$	0.14	0.45	2.2
+ decoupled-GRPO	$+0.292$	0.41	$+0.040$	0.36	0.16	4.3

5.7 Summary of the empirical findings

Across the experiments, no single RL method dominates across clinical severity, interaction burden, and task structure. Outcome only GRPO frequently improves aggregate value by shortening the dialogue, but this can reduce high risk reliability. Credit assignment produces the strongest replicated improvement in diagnostics; it raises held out value and removes the observed high risk seed collapse.

Decoupled exploration further improves high risk support, but imposes additional burden on lower acuity cases. In titration, the benefit of information credit depends on its scale and on the limited adaptive headroom of the task.

The SFT results place a further boundary on the value of RL. When supervised training already captures the desired procedure, RL may have little useful behaviour left to improve and can instead erode the initialization. When SFT contains a rare coverage gap, the ability to sample the missing action does not ensure that RL will repair it.

The main empirical conclusion is therefore conditional. Reinforcement learning can improve clinical information gathering when the task contains adaptive headroom, the initial policy exposes useful trajectories, and the reward assigns credit to the evidence gathering process. When these conditions are absent or poorly calibrated, RL can replace one failure mode with another while preserving or even improving the aggregate score.

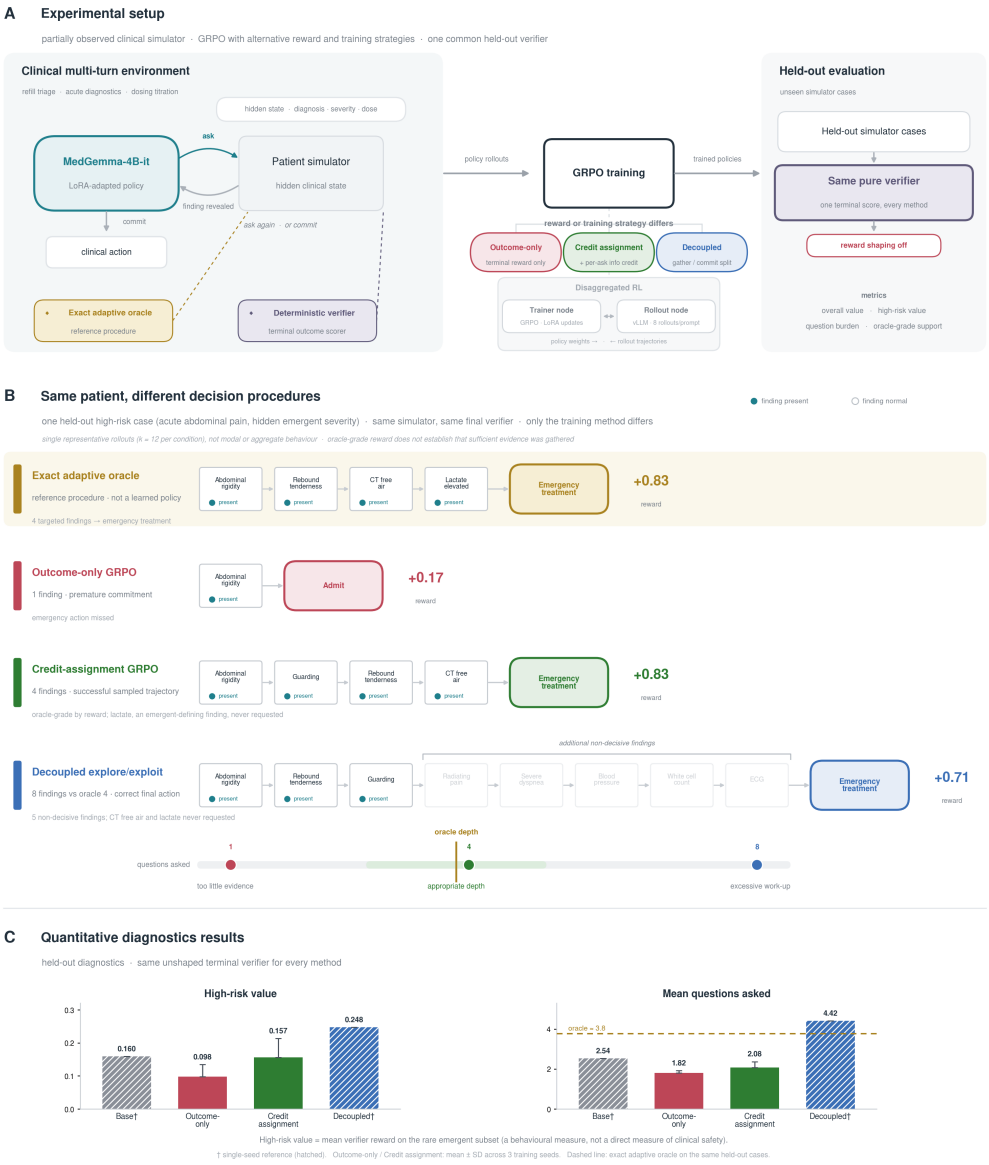


Figure 1: Experimental framework and effects of reward design on clinical multi-turn decision procedures. **A,** The LoRA-adapted MedGemma-4B-it policy interacts with a partially observed patient simulator across medication-refill triage, acute diagnostics and dosing titration. Each environment provides an exact adaptive oracle and deterministic outcome verifier. The three GRPO methods use different training reward formulations or exploration strategies, with policy optimization and rollout generation distributed across two compute nodes. All policies are evaluated on held-out cases using the same unshaped terminal verifier. **B,** Selected trajectories for the same simulated high-risk abdominal-pain case illustrate different information-gathering procedures. The exact adaptive oracle gathers four targeted findings before emergency treatment. Outcome-only GRPO commits prematurely after one finding. Credit-assignment GRPO reaches the correct terminal action after four requests and receives an oracle-grade reward despite following a different evidence sequence. Decoupled exploration reaches the correct terminal action after eight requests, including additional non-decisive findings. The learned trajectories are selected illustrative examples, not representative estimates of typical policy behavior. Oracle-grade reward does not necessarily establish sufficient gathered clinical evidence. **C,** Held-out diagnostics performance measured by high-risk verifier value and mean questions asked. Outcome-only and credit-assignment GRPO show mean ± standard deviation across three training seeds; the base model and decoupled GRPO are single-seed references (hatched bars). The dashed line indicates the exact adaptive oracle's mean question count. All methods are scored using the same unshaped evaluation reward.

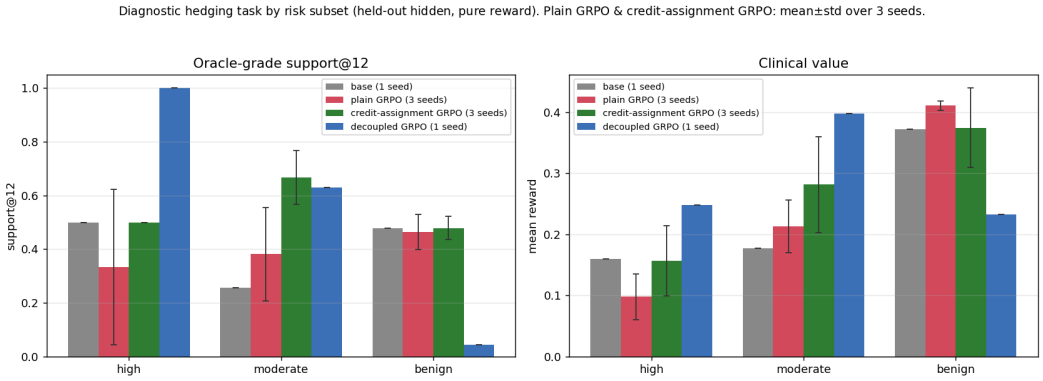


Figure 2: Diagnostics performance by risk subgroup in the held out hidden condition. Every method is evaluated with the pure verifier reward. Plain GRPO and credit assignment GRPO show mean±standard deviation over three training seeds; base and decoupled GRPO are single seed references. Outcome only GRPO raises aggregate value but reduces and destabilizes high risk performance. Credit assignment improves the moderate subgroup and maintains high risk support of 0.50 across all three seeds. Decoupled GRPO produces the highest high risk value and support, but requests substantially more information.

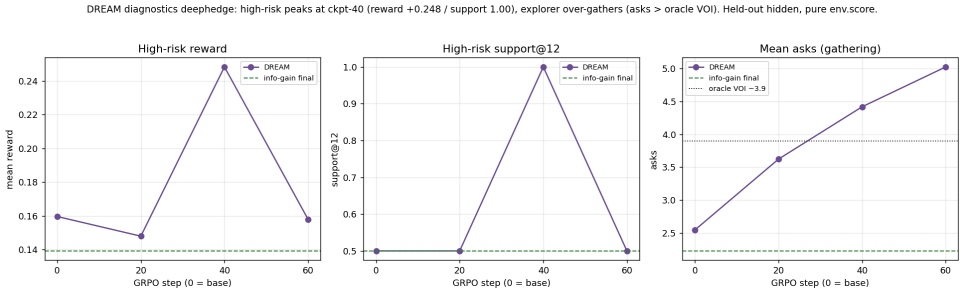


Figure 3: Checkpoint trajectory for decoupled GRPO on diagnostics. Evaluation uses the held out hidden condition and the pure verifier reward. High risk oracle grade support and high risk value peak at checkpoint 40, reaching 1.00 and +0.248, respectively. The same checkpoint also shows increased gathering; mean questions approach and then exceed the oracle level of approximately 3.9. The trajectory illustrates the method's central tradeoff between high risk coverage and information burden.

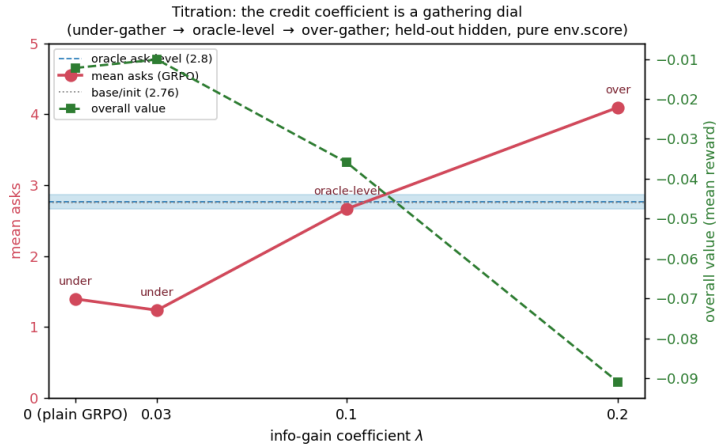


Figure 4: Information credit coefficient sweep on the titration task. Mean questions remain below the oracle level when $\lambda \leq 0.03$, approach the oracle at $\lambda = 0.1$, and exceed it at $\lambda = 0.2$. The coefficient therefore behaves as a task specific control parameter for gathering depth, rather than as a constant that can be transferred without calibration.

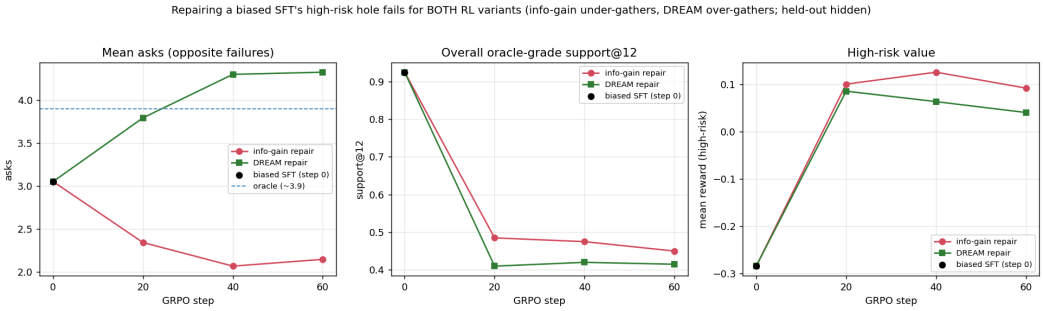


Figure 5: Attempted reinforcement learning repair of the biased SFT policy. Credit assignment and decoupled GRPO move away from the initialization in opposite directions. Credit assignment reduces gathering and loses high risk support; decoupled GRPO increases gathering and damages benign support. Both methods reduce the SFT policy's strong aggregate support, and neither restores the missing high risk procedure.

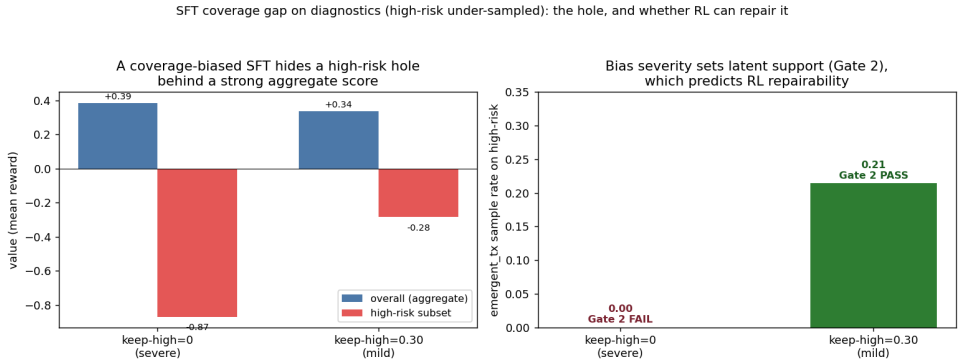


Figure 6: Effect of supervised coverage bias on aggregate performance and latent support. Left, the biased SFT retains high overall support despite poor high risk value. Right, severe bias removes the high risk action `emergent_tx` from sampled trajectories and fails Gate 2. Mild bias retains a 21% sampling rate and passes Gate 2. The subsequent repair experiment shows that nonzero latent support is necessary but not sufficient for RL to restore the missing behaviour.

6 Discussion

The central conclusion is not that reinforcement learning consistently improves clinical multi turn agents. RL can improve, preserve, or degrade the decision procedure depending on the task, the initial policy, and the structure of the reward. The strongest positive result appears in diagnostics, where credit assignment improves held out value and stabilizes high risk support relative to outcome only GRPO across three seeds. The remaining experiments define important limits. Stronger exploration improves high risk coverage but increases question burden; calibrated gathering yields little additional value when the task has limited adaptive headroom; and RL does not reliably improve a supervised policy that is already strong or repair one with a rare coverage gap.

These findings suggest that the important question is not simply whether RL raises average reward. The more useful question is whether RL teaches a reliable procedure for gathering evidence and committing to an action. In the present environments, those two outcomes are not equivalent. A policy can improve its mean score by asking fewer questions, choosing a broadly aggressive action, or using an intermediate decision that performs adequately on common cases. Such a policy may optimise the verifier without learning the clinical process that the verifier was intended to encourage.

A practical decision rule for RL post training. The experiments support a staged approach to reinforcement learning for clinical agents. First, the task should be tested for adaptive headroom. When the best fixed action performs nearly as well as the exact adaptive oracle, there is little decision value for RL to recover through information gathering. Second, the initial policy should be tested for latent behavioural support. If it never samples the desired gather then commit trajectory, policy optimisation has no successful behaviour to amplify. Third, when both gates pass, the reward should assign credit to clinically useful intermediate decisions rather than relying only on the final action.

These gates are negative screens, not certificates of safety. Passing them shows that RL has a plausible opportunity to help; it does not show that outcome only GRPO will use that opportunity appropriately. Diagnostics passes both gates, yet plain GRPO improves aggregate value while reducing high risk performance and question depth. The reward structure remains the controllable factor. Credit assignment improves the replicated diagnostics comparison, while decoupled exploration shifts the policy toward greater high risk coverage at the cost of over gathering. The titration sweep further shows that the strength of process credit must be calibrated to the task.

Supervised initialization adds a fourth consideration. If SFT has already learned most of the behaviour measured by the verifier, RL may have little useful headroom. If the SFT policy contains a rare coverage gap, the presence of the missing action in occasional stochastic rollouts does not ensure that standard RL will repair the associated procedure. A practical pipeline should therefore measure both SFT performance and residual error structure before deciding whether additional RL is justified.

The safety question concerns procedure, not only outcome. Clinical decisions are usually justified by evidence. A medication refill may be appropriate only after checking renal function, contraindications, or adverse effects. Emergency treatment may be appropriate only after obtaining findings that support a high acuity state. Two agents can select the same final action while differing substantially in reliability; one may have gathered the necessary evidence, while the other may have guessed correctly or followed a fixed action bias.

This distinction explains why average reward and final action accuracy are insufficient. In diagnostics, plain GRPO improves overall value relative to the base model, but high risk value and support decline. In refill triage, support at twelve sampled rollouts remains high after RL, yet the typical policy asks far fewer questions than the supervised initialization. The guarded analysis therefore classifies the apparent support as an artefact of stochastic sampling around an under gathering policy, rather than evidence of a reliable clinical procedure.

Clinical evaluation should consequently include both outcome and process measures. Relevant audits include whether the model obtained the findings needed to justify its decision, whether it committed prematurely, whether it repeatedly used a convenient intermediate action, whether it escalated without support, and whether it gathered excessive information on benign cases. Rare high risk cases should be reported separately because aggregate metrics can remain stable while performance on the most consequential subgroup deteriorates.

Implications for clinical decision support. The present environments are not deployment studies, but the failure modes have direct relevance to future clinical decision support systems. A model used for refill review could reduce interaction burden by asking fewer questions while silently omitting a renal or contraindication check. An acute assessment agent could appear sensitive to emergencies because it escalates broadly, even when its decisions are unsupported by the available evidence. A system designed to avoid those misses could fail in the opposite direction by requesting every available finding and creating unnecessary delay or workload.

These behaviours matter because clinical safety is not defined by sensitivity alone. A useful system must balance missed danger, unnecessary escalation, information burden, and the strength of evidence supporting its recommendation. The results show that reward design can move the policy among these operating points without producing the intended balance. A higher verifier score should therefore not be interpreted as a clinically meaningful improvement unless the trajectory analysis shows that the model gathered and used the appropriate evidence.

The findings also have implications for supervision. A clinician reviewing only the final recommendation may not detect that the policy reached it through an unstable procedure. Supervision tools should expose which findings were requested, which evidence changed the action, what remained unknown at commitment, and whether the decision would have changed under plausible alternate findings. Such process visibility may be as important as the final confidence score.

Why clinical information gathering differs from math and code RLVR. RLVR has produced strong results in domains where correctness can be sharply verified, including mathematics and programming [Shao et al., 2024]. In those settings, an incorrect answer or failing program usually receives an immediate terminal penalty. Clinical information gathering contains an additional layer. The desired output is not only an action; it is a sequence of decisions under uncertainty about which evidence to obtain, how much evidence is sufficient, and when the cost of further investigation exceeds its expected value.

This structure creates more opportunities for shortcut optimisation. A policy may avoid the difficult high risk decision by selecting an intermediate action. It may ask one highly predictive question and ignore the remaining evidence needed for a robust decision. It may also stop before obtaining information that would expose an error in its preferred action. These procedures can receive acceptable average reward even though they are not reliable clinical strategies.

The implication is not that RLVR is unsuitable for clinical agents. Rather, the verifiable objective must represent more of the procedure. Credit assignment is one approach because it rewards questions that reduce uncertainty about the hidden state. Decoupling exploration from exploitation is another because it gives information gathering its own optimisation signal. The tradeoffs observed here show that neither mechanism is automatically safe; each changes the balance between gathering, commitment, and burden.

SFT changes the interpretation of RL gains. Most practical clinical models will undergo supervised adaptation before reinforcement learning. The incremental question is therefore whether RL improves what SFT leaves unresolved, rather than whether it can teach the task from an unadapted base model.

The refill experiment illustrates the limited headroom case. The SFT warmup already learns much of the medication safety procedure and reaches strong high risk support. Continued RL changes the model, but the dominant change is toward fewer questions and lower high risk value. A small or negative incremental gain in this setting does not establish that RL is generally ineffective; it indicates that the supervised model had already captured much of the measurable benefit and that the remaining reward pressure favoured a cheaper procedure.

The biased diagnostics experiment illustrates the opposite case. A supervised policy can perform strongly on average while retaining a rare high risk weakness. RL is an appealing repair mechanism, but repair is not automatic. When the high risk action is absent from the policy distribution, there is no useful trajectory to reinforce. When the action is present but rare, both tested RL methods move toward easier neighbouring strategies rather than selectively restoring the missing behaviour. Credit assignment under gathers; decoupled GRPO over gathers and damages benign performance.

These results suggest that supervised coverage remains a primary safety concern. RL can amplify behaviour that is already represented and rewarded frequently enough, but it should not be assumed to discover every omitted clinical pattern. Targeted supervised examples, risk balanced sampling, explicit subgroup rewards, or curriculum design may be required before RL can provide a reliable incremental gain.

Relation to clinical RLHF and preference optimisation. The experiments use analytic verifiers because they make the state, utility, and optimal adaptive policy measurable. This does not imply that every clinically important objective can be reduced to a deterministic reward. Communication quality, calibrated uncertainty, empathy, shared decision making, and clinician or patient preferences often require human judgement. RLHF, direct preference optimisation, and related methods are more natural for these dimensions.

The cautionary result nevertheless extends beyond analytic RLVR. A policy that can exploit a deterministic verifier may also exploit a learned reward model. Learned feedback can introduce additional vulnerabilities, including inconsistency, supervisor bias, excessive caution, or preference for stylistically reassuring responses over well supported decisions. Process level and subgroup evaluation remain necessary regardless of whether the reward is analytic or learned.

A future clinical alignment pipeline will probably be hybrid. Deterministic verification may be appropriate for components such as dose calculations, contraindication checks, interaction checks, monitoring requirements, schema validity, and minimum information sufficiency. Preference based methods may be better suited to communication, uncertainty expression, and preference sensitive tradeoffs. Both components should be evaluated using the same guarded principle: the final response is not sufficient evidence that the underlying procedure is reliable.

Limitations. The experiments use synthetic, stylized environments with deterministic patient responses and hand specified utility matrices. They are designed to isolate learning dynamics, not to estimate real clinical benefit or harm. The cases do not include noisy histories, ambiguous language, documentation errors, uncertain allergy records, conflicting laboratory results, changing patient goals, or clinician disagreement. The utility matrices encode experimental assumptions about relative consequences and have not been validated by clinical experts. The trained policies should therefore not be interpreted as suitable for patient care.

The evidence is also limited to one model family and scale, `medgemma-4b-it`, with bounded LoRA based training runs. The main diagnostics comparison between outcome only GRPO and credit assignment is replicated across three seeds. The decoupled method, titration coefficient sweep, refill experiments, and SFT repair experiments remain single seed. Conclusions from those cells are therefore exploratory, particularly where checkpoint selection or stochastic support metrics are involved.

Support@12 is useful for measuring whether a behaviour exists within the policy distribution, but it is not a direct estimate of single trajectory deployment safety. A policy can obtain high support because one of twelve samples succeeds while the majority follow an unsafe procedure. Future studies should place greater emphasis on deterministic or greedy evaluation, pass at one performance, premature commitment rates, supported versus unsupported escalation, expected regret, and subgroup specific uncertainty intervals.

The SFT analysis is also incomplete. I evaluate one light protocol warmup and one family of biased diagnostics initializations, but I do not measure a complete dose curve across SFT strength and coverage. I therefore show that strong SFT can leave little RL headroom and that RL can fail to repair a biased policy; I do not characterize every supervised regime or every possible repair method.

Finally, the number of high risk cases is small in the principal held out sets. Reporting the subgroup separately prevents it from being hidden by the aggregate, but larger stratified evaluations would produce more precise estimates. Future work should use paired case level comparisons, confidence intervals, multiple training seeds, and larger high risk test sets before drawing stronger safety conclusions.

Future work. The most direct next experiment is a controlled SFT strength and coverage study. Models should be trained on increasing amounts of well covered supervised data, then evaluated before and after matched RL. In parallel, high risk coverage should be varied while total SFT quality

is held approximately constant. This would identify when RL has genuine residual headroom, when it can repair a rare weakness, and when it merely reshapes an already strong policy.

A second direction is improved reward design. Entropy reduction is useful because it is analytic, but it rewards uncertainty reduction rather than the decision value of a question. Future rewards could estimate expected regret reduction, probability of changing the optimal action, or satisfaction of explicit evidence requirements. These alternatives may distinguish clinically decisive questions from findings that reduce uncertainty without affecting management.

A third direction is more realistic interaction. Future environments should include stochastic patient responses, ambiguous findings, conflicting chart information, missing laboratory values, uncertain medication histories, and multiple clinically acceptable stopping points. Expert reviewed utilities and evidence requirements would be needed before the benchmark could support stronger clinical claims.

A fourth direction is goal conditioned decision making. The same patient state may require a different operating point depending on whether the objective is to minimise missed emergencies, reduce unnecessary escalation, limit patient burden, or prepare a clinician handoff. Representing these goals explicitly would allow the policy to learn calibrated tradeoffs rather than relying on one fixed utility matrix.

The long term goal is a reusable benchmark for clinical multi turn agents in which RL is evaluated as procedure learning. Such a benchmark should measure whether the agent requests the findings that matter, stops when enough is known, commits to an action supported by the available evidence, and preserves safety on rare high consequence cases. Aggregate reward should remain part of the evaluation, but it should not be the sole criterion for selecting a clinical policy.

Conclusion. The experiments show that verifiable outcomes alone do not ensure reliable clinical information gathering. Outcome only GRPO can improve average reward while reducing evidence gathering and weakening high risk behaviour. Dense credit assignment provides a meaningful improvement in the replicated diagnostics comparison, but its benefit depends on task headroom and reward calibration. Stronger exploration can improve high risk coverage while increasing burden, and RL does not reliably add value beyond a strong SFT policy or repair rare omissions in biased supervised data.

The broader implication is that reinforcement learning for clinical agents should be treated as optimisation of a sequential decision procedure, not merely a final answer. Safe post training requires adaptive headroom, behavioural support in the initial policy, appropriate intermediate credit, and evaluation that exposes rare procedural failures rather than averaging them away.

Addendum: additional refill triage and coverage results

The following figures provide additional subgroup results for the refill triage and biased SFT experiments. They complement the main results by showing how aggregate support can conceal deterioration in high risk behaviour or excessive information gathering.

A Implementation details

All experiments adapt the text only `medgemma-4b-it` base model using LoRA with rank 32 and α=64. Tables 6 and 7 report the supervised and reinforcement learning hyperparameters.

Training uses a disaggregated two node system. One NVIDIA DGX Spark node performs model optimisation, while a second node runs a vLLM server that generates on policy trajectories. Both systems use Grace Hopper hardware with unified memory. A supervised warmup requires approximately 0.5 hours, while a 60 step GRPO run generally requires 6 to 8 hours.

Figure 9 summarizes the principal optimisation diagnostics. The upper panels show reward, divergence from the initialization, policy entropy, and mean questions per rollout for the diagnostics experiments. The lower panel shows the loss curve for the biased SFT warmup. All values are read directly from the corresponding training logs.

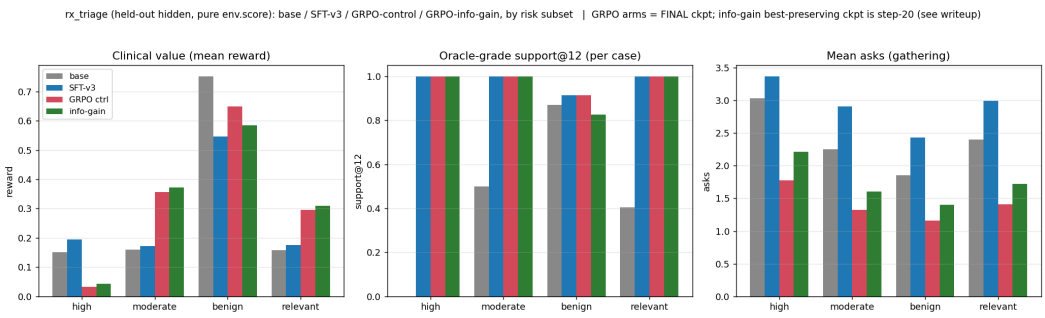


Figure 7: Refill triage performance by risk subgroup in the held out hidden condition, evaluated with the pure verifier reward. The figure compares the base model, the SFT v3 warmup, outcome only GRPO, and credit assignment GRPO at their final checkpoints. The SFT warmup already achieves the strongest high risk result. Both RL policies subsequently reduce high risk value and move toward shorter, under gathered trajectories, despite preserving high support under stochastic sampling.

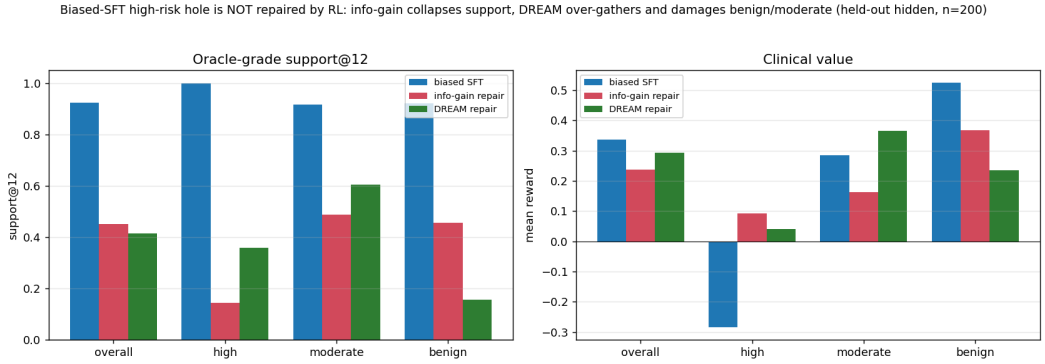


Figure 8: Attempted repair of a biased diagnostics SFT policy by risk subgroup, using the held out hidden condition and n=200 cases. Neither RL method restores the original high risk oracle grade support; support falls from 1.00 for the SFT initialization to 0.14 with credit assignment GRPO and 0.36 with decoupled GRPO. The methods fail in different directions. Credit assignment reduces gathering, while decoupled GRPO increases question burden and lowers benign support from 0.92 to 0.16. This figure complements Table 5 and Figure 5.

B Representative oracle trajectories

The following examples illustrate the common ask to commit structure used across the three environments. Each trace shows one held out case and the corresponding exact adaptive oracle trajectory. The simulator reveals only findings that the agent explicitly requests; all other findings remain hidden.

These examples are intended to clarify the experimental interface and scoring procedure. They are simplified simulator traces rather than realistic clinical conversations. Future versions should include richer successful and failed trajectories, including premature commitment, unsupported escalation, hedging, and unnecessary information gathering.

```
DIAGNOSTICS (deephedge) -- true latent: high (emergency); oracle: emergent_tx
  user:      Presentation: adult with acute chest pain. Chart: no results yet.
  assistant: {"action_type":"ask","intent":"troponin_high"}   user: troponin elevated
  assistant: {"action_type":"ask","intent":"ecg_ischemia"}    user: ECG shows ischemia
  assistant: {"action_type":"ask","intent":"dyspnea_severe"}  user: severe dyspnea
  assistant: {"action_type":"ask","intent":"hypotension"}     user: hypotensive
  assistant: {"action_type":"commit","decision":"emergent_tx",...}   reward = 0.83

REFILL TRIAGE -- true latent: moderate; oracle: dose_adjust
  user:      Refill request: ACE inhibitor / ARB (lisinopril). Chart: no recent labs.
  assistant: {"action_type":"ask","intent":"renal_decline"}     user: eGFR declining
  assistant: {"action_type":"ask","intent":"egfr_severe"}       user: decline mild
```

Table 6: Hyperparameters for the supervised protocol warmup. Loss is computed on every assistant turn; system messages, user messages, and simulator responses are excluded from the supervised objective.

Base model	`medgemma-4b-it` (text-only causal LM)
Adapter	LoRA r=32, α=64, dropout 0.05, all-linear targets
Objective	causal-LM cross-entropy on *every* assistant turn (system/user masked)
Optimizer	AdamW, lr 1×10^{-4}, linear warmup 5% then constant
Epochs / batch / max seq len	2 / 4 / 2048
Training trajectories	800 (oracle-adaptive; generation seed 7)
Seed / checkpoint	1 / `final`

Table 7: Hyperparameters for GRPO training. Outcome only GRPO uses the terminal utility minus the accumulated question cost. Credit assignment adds $\lambda\Phi$ to the rollout reward during training only. Decoupled GRPO instead alternates exploration updates based on state recoverability with exploitation updates based on `env.score`. Every policy is evaluated using the same unshaped terminal verifier.

Algorithm	GRPO (TRL), custom on-policy `rollout_func`
Initialization	base model, or merged SFT adapter
Adapter	LoRA r=32, α=64 (q,k,v,o,gate,up,down)
Group size (rollouts/prompt)	8
Per-device batch / grad-accum	8 / 4
Training cases / steps / save	300 / 60 / every 20
Learning rate / KL β	1×10^{-5} / 0.02
Rollout temperature / max tokens-per-turn	1.0 / 128
Max completion length / turn cap	512 / 10–12
Importance-sampling correction	off (fully on-policy, ratio ≈ 1)
EOS token ids / token masking	$\{1, 106\}$ / `env_mask` (model tokens only)
Reward	$U[\mu][a] - 0.03\, n_{\text{asks}}$ ($+\lambda\Phi$ for credit assignment)
Credit assignment λ	0.2 (diagnostics/refill); swept $\{0, 0.03, 0.1, 0.2\}$ (titration)
Seed	1

```
  assistant: {"action_type":"ask","intent":"volume_dehydrated"} user: dehydrated
  assistant: {"action_type":"commit","decision":"dose_adjust",...}   reward = 0.81

TITRATION -- true latent: high clot / low bleed; oracle: high (full intensity)
  user:      Patient: older adult with atrial fibrillation. Chart: none documented.
  assistant: {"action_type":"ask","intent":"bleed_history"}  user: no prior bleeding
  assistant: {"action_type":"ask","intent":"fall_risk"}      user: low fall risk
  assistant: {"action_type":"commit","decision":"high",...}   reward = 1.34
```

The diagnostics trajectory obtains four findings before selecting emergency treatment. Its terminal utility is reduced by four question costs, producing the reported reward of 0.83. The refill trajectory obtains three findings before modifying the medication dose, resulting in reward 0.81. The titration trajectory asks about two bleeding risk factors before selecting full treatment intensity.

The examples demonstrate what an oracle trajectory looks like, but they do not show the failure modes that distinguish the trained policies. A more complete qualitative appendix should place each oracle trace beside at least one successful model trajectory and one failed trajectory. The failed examples should include the corresponding terminal utility, question cost, process classification, and the reason that the trajectory is considered premature, unsupported, hedged, or excessive.

Code

`https://github.com/csbrendan/rl-gains-in-clinical-tasks`

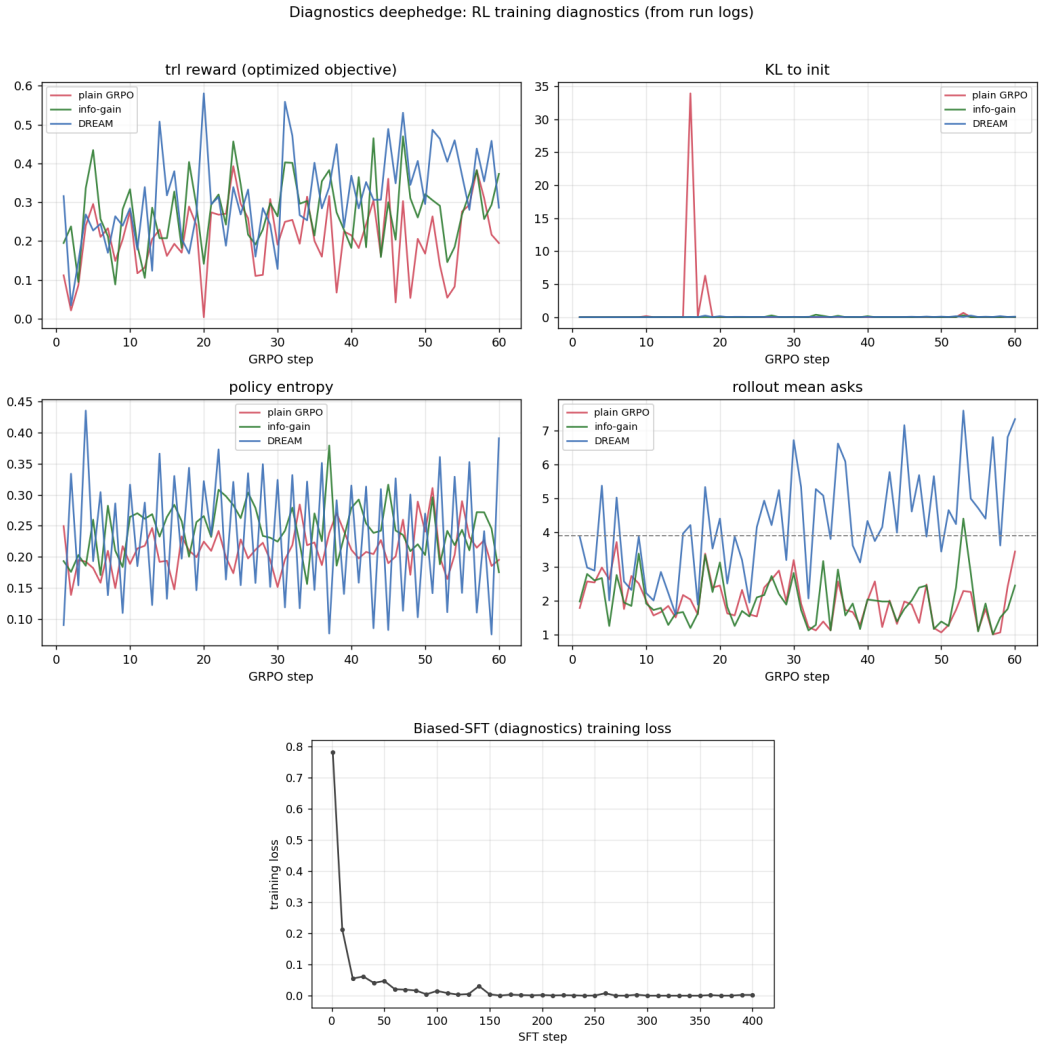


Figure 9: Training diagnostics derived from the run logs. Top, diagnostics GRPO over 60 optimisation steps. The panels show the optimised reward, KL divergence from the initialization, policy entropy, and mean questions per rollout for outcome only GRPO, credit assignment GRPO, and decoupled GRPO. The question trajectory reveals the characteristic procedural differences among the methods; outcome only GRPO moves below the oracle gathering level of approximately 3.9, credit assignment preserves an intermediate level, and decoupled GRPO gathers the most. Bottom, training loss for the biased SFT warmup. Loss converges to approximately 0.002, showing that the supervised trajectories are closely fit before RL begins.

References

Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling, 2024. URL `https://arxiv.org/abs/2407.21787`.

Junying Chen, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, Jianye Hou, and Benyou Wang. HuatuoGPT-o1: Towards medical complex reasoning with LLMs. *arXiv preprint arXiv:2412.18925*, 2024.

Chengfeng Dou, Ying Zhang, Zhi Jin, Wenpin Jiao, Haiyan Zhao, Yongqiang Zhao, and Zhengwei Tao. Integrating physician diagnostic logic into large language models: Preference learning from process feedback. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2453–2473. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.findings-acl.144. URL `https://aclanthology.org/2024.findings-acl.144/`.

Yichun Feng, Jiawei Wang, Lu Zhou, Yikai Zheng, Zhen Lei, and Yixue Li. Real-world doctor agent with proactive consultation through multi-agent reinforcement learning, 2026.

Anna Goldie, Azalia Mirhoseini, Hao Zhou, Irene Cai, and Christopher D. Manning. Synthetic data generation & multi-step rl for reasoning & tool use, 2025. URL https://arxiv.org/abs/2504.04736.

Jubayer Ibn Hamid, Ifdita Hasan Orney, Michael Y. Li, Omar Shaikh, Yoonho Lee, Dorsa Sadigh, Chelsea Finn, and Noah Goodman. Spiral: Learning to search and aggregate, 2026. Preprint, ongoing work.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step, 2023. URL https://arxiv.org/abs/2305.20050.

Sicheng Lin, Yifan Wu, and Jimeng Sun. Training LLMs for EHR-based reasoning tasks via reinforcement learning (EHRMIND). *arXiv preprint*, 2025.

Evan Zheran Liu, Aditi Raghunathan, Percy Liang, and Chelsea Finn. Decoupling exploration and exploitation for meta-reinforcement learning without sacrifices. In *International Conference on Machine Learning (ICML)*, 2021.

Andrew Y. Ng, Daishi Harada, and Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *International Conference on Machine Learning (ICML)*, 1999.

Harsha Nori, Mayank Daswani, Christopher Kelly, Scott Lundberg, Marco Tulio Ribeiro, Marc Wilson, Xiaoxuan Liu, Viknesh Sounderajah, Jonathan Carlson, Matthew P. Lungren, Bay Gross, Peter Hames, Mustafa Suleyman, Dominic King, and Eric Horvitz. Sequential diagnosis with language models, 2025. URL https://arxiv.org/abs/2506.22405.

Khaled Saab, Robert Carter, Aidan Hong, et al. MedGemma technical report. *arXiv preprint arXiv:2507.05201*, 2025.

Thomas Savage, Stephen P. Ma, Abdessalem Boukil, Ekanath Rangan, Vishwesh Patel, Ivan Lopez, and Jonathan Chen. Fine-tuning methods for large language models in clinical medicine by supervised fine-tuning and direct preference optimization: Comparative evaluation. *Journal of Medical Internet Research*, 27:e76048, 2025. doi: 10.2196/76048. URL https://www.jmir.org/2025/1/e76048/.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

Jonathan Uesato, Nate Kushman, Ramana Kumar, Francis Song, Noah Siegel, Lisa Wang, Antonia Creswell, Geoffrey Irving, and Irina Higgins. Solving math word problems with process- and outcome-based feedback, 2022. URL https://arxiv.org/abs/2211.14275.

Huaijie Wang, Shibo Hao, Hanze Dong, Shenao Zhang, Yilin Bao, Ziran Yang, and Yi Wu. Offline reinforcement learning for LLM multi-step reasoning. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 8881–8893. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-acl.464. URL https://aclanthology.org/2025.findings-acl.464/.

Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9426–9439. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.510. URL https://aclanthology.org/2024.acl-long.510/.

Sheng Zhang et al. MED-RLVR: Emerging medical reasoning from a 3B base model via reinforcement learning. *arXiv preprint arXiv:2502.19655*, 2025.

Siyan Zhao, Zhihui Xie, Mengchen Liu, Jing Huang, Guan Pang, Feiyu Chen, and Aditya Grover. Self-distilled reasoner: On-policy self-distillation for large language models, 2026.